

Julio César Sandoval

Generative AI Architect · AI Engineer · GenAI Solutions Consultant

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PROFILE

Software engineer and data scientist with **4 years building enterprise software** and **3+ years shipping generative AI to production**. Designs multi-agent and RAG systems on Microsoft Azure that put financial and operational answers in front of C-level users. Works end to end — requirements, architecture, Python and SQL implementation, deployment — and collaborates directly with Microsoft on enterprise delivery.

TECHNICAL EXPERTISE

Generative AI	Agentic & multi-agent systems · Agentic RAG · Advanced prompting techniques · Context optimization · Embeddings · Evaluation, testing and performance metrics · Guardrails and LLM vulnerability mitigation
Models & Frameworks	GPT · Claude · Model Context Protocol (MCP) · Claude Code, Github Copilot and AI-assisted workflows
Data & Backend	Python · SQL · Vector databases
Cloud & Delivery	Microsoft Azure · Google Cloud · Git
Software Engineering	.NET · Angular

EXPERIENCE

Data Scientist, Generative AI — Derevo March 2024 – Present
Remote, Mexico · Microsoft partner · enterprise data & AI consultancy

- Architect and ship **multilingual multi-agent chatbots** that give C-level executives self-service access to financial and operational data through Microsoft Teams and Web.
- Led AI development of **LUCA Bot for Cemex México**, delivered with Derevo and Microsoft: a multi-agent system on Azure and Copilot Studio giving executives secure, instant access to business information.
- Built **C-Agent**, Derevo's flagship GenAI decision-support product, exhibited at Microsoft's Innovation Hub in Mexico City.
- Designed **RAG pipelines** that query enterprise SQL databases holding millions of records alongside thousands of unstructured PDFs, generating SQL and Python from natural language and rendering live visualizations.
- Collaborate **directly with Microsoft** on architecture, security and deployment; translate client requirements into production solutions, lead small delivery teams and train client and internal teams in prompt engineering.

STACK Python · Azure OpenAI · SQL · LLMOps

Data Scientist — AARFS A.C. January 2023 – December 2023
Los Mochis, Sinaloa, Mexico

- Built **GPT-3.5 chatbots** for technical support and retrieval of SQL and ERP documentation, aimed at reducing repetitive support tickets.
- Developed and deployed **sales forecasting models** with RNN and LSTM architectures in TensorFlow, from data preparation to production integration with SQL Server.
- Integrated AI capabilities into the company ERP using Angular, C# and SQL Server.

Fullstack Developer — Finesoft June 2022 – December 2022
Los Mochis, Sinaloa, Mexico

- Developed and maintained web billing tools with C#, ASP.NET and SQL Server; improved responsive interfaces with Bootstrap and CSS.

CERTIFICATIONS

- GitHub Copilot · Microsoft · Mar 2026 – Mar 2028
- Databricks Certified Generative AI Engineer Associate · Dec 2025 – Dec 2027
- Agentic AI Proficient Developer, FY25 · Microsoft · Jul 2025

SELECTED PROJECTS

- Prompt Engineering: A Practical Guide** — 23-page manuscript with applied guidance and GPT-4o examples.
- Castube** — natural-language YouTube search with timestamped results and summaries (Angular, OpenAI).
- HTMLGen** — conversational generator of functional HTML / CSS / JS components (Angular, OpenAI).
- Face generation fine-tuning** — LoRA fine-tune of Flux-dev with ComfyUI workflows.

EDUCATION & LANGUAGES

B.Sc. Software Engineering — Universidad Autónoma de Sinaloa · 2017–2022 · 1st place, Coding Cup Sinaloa (ICPC "Ada Lovelace") 2019 · EGEL+ outstanding in Software Development

Spanish native · English B1